# UniLR: Unleashing the Power of LLMs on Multiple Legal Tasks with a Unified Legal Retriever

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## Abstract

Despite the impressive capabilities of LLMs, they often generate content with factual inaccuracies in LegalAI, which may lead to serious legal consequences. Retrieval-Augmented Generation (RAG), a promising approach, can conveniently integrate specialized knowledge into LLMs. In practice, there are diverse legal knowledge retrieval demands (e.g. law articles and similar cases). However, existing retrieval methods are either designed for general domains, struggling with legal knowledge, or tailored for specific legal tasks, unable to handle diverse legal knowledge types. Therefore, we propose a novel **Uni**fied **L**egal **R**etriever (UniLR) capable of performing multiple legal retrieval tasks for LLMs. Specifically, we introduce attention supervision to guide the retriever in focusing on key elements during knowledge encoding. Next, we design a graph-based method to integrate meta information through a heterogeneous graph, further enriching the knowledge representation. These two components work together to enable UniLR to capture the essence of knowledge hidden beneath formats. Extensive experiments on multiple datasets of common legal tasks demonstrate that UniLR achieves the best retrieval performance and can significantly enhance the performance of LLM.

## 1 Introduction

Legal artificial intelligence (LegalAI) (Zhong et al., 2020a) focuses on applying artificial intelligence to benefit legal tasks (Zhong et al., 2018, 2020b; Wu et al., 2020, 2022). Recently, the focus has shifted towards leveraging large language models (LLMs) to enhance legal task performance (Fei et al., 2024). However, LLMs still struggle with factual inaccuracies (Mallen et al., 2023; Min et al., 2023), which can lead to serious legal consequences. A promising solution is Retrieval-Augmented Generation (RAG) (Ram et al., 2023), where a retriever fetches relevant knowledge from an external corpus and combines it with the query to generate a more accurate response.

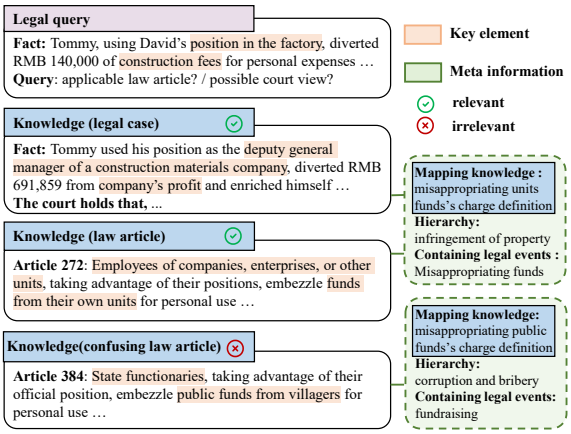



Figure 1: A real-world legal example. For a fact description, if the query asks for applicable articles, relevant articles should be retrieved; if it asks for a possible court view, similar cases should be retrieved for reference.

Legal knowledge has various types such as charge definition, law articles, similar cases, and so on (Burton, 2007). As shown in Fig. 1, in LegalAI practice, addressing various legal tasks requires retrieving different types of legal knowledge. Generally, existing retrieval methods can be divided into two types: Sparse retrieval, like BM25 (Robertson and Zaragoza, 2009) and TF-IDF (Sparck Jones, 1972), can be applied to multiple tasks but struggle with processing complex legal knowledge. Dense retrieval, while effective for specific legal tasks through fine-tuning, cannot handle diverse retrieval needs (Wang et al., 2018, 2019a; Li et al., 2023a). Therefore, it is meaningful to develop a unified retriever that can address multiple legal retrieval tasks.

However, the main challenge in implementing such a retriever lies in the legal knowledge format

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problem. As shown in Fig. 1, the challenge manifests in two aspects: (1) **Diverse formats across different knowledge types.** For example, law articles are filled with concise legal terminology, while relevant cases are lengthy fact descriptions, which hinders the retriever's understanding. (2) **Subtle differences within the same knowledge type.** In the legal domain, differing only in a few words can yield markedly different legal implications, which may trigger the retriever's confusion. Notably, our findings show that, for the query-relevant knowledge, despite format varying, **key elements**[1] (e.g., defendant's identity, behavior's target) are similar, and **meta information**[1] (e.g., mapping knowledge, hierarchy, legal event schema) remains consistent. Irrelevant but confusing knowledge can also be distinguished using these concepts.

Based on this insight, we propose UniLR, a novel **Uni**fied **L**egal **R**etriever for handling multiple legal retrieval tasks by leveraging key elements and meta information. First, we develop a knowledge mining process. For key elements, we distill expertise from a legal model (Yao et al., 2022), using its attention distribution to identify key elements. For meta information, we construct a heterogeneous knowledge graph where legal knowledge, hierarchical structures, and legal event schemas are represented as different node types, with inter-type and cross-type relationships established based on carefully crafted edge construction rules.

To leverage the mined key elements and meta information, based on a dense embedding model, we design two innovative components: Key Element Supervisor (KES) and Graph-based Knowledge Augmenter (GKA). KES leverages recorded attention and a supervision loss to align the model's focus on key elements. GKA combines graph attention and convolution to aggregate meta information from the graph, enhancing knowledge representation. Finally, we train UniLR using contrastive learning.

Multiple datasets for common legal tasks are experimented in this paper, including charge prediction, law article prediction, court view generation, and legal question answering. Empirical results demonstrate that UniLR achieves the best retrieval performance and significantly enhances LLM's performance. To summarize, our contributions are:

- We investigate multiple legal retrieval tasks for RAG in the LLM era, considering the diverse formats and knowledge confusion.
- We define key elements and meta information of legal knowledge and mine them through expertise distillation and heterogeneous knowledge graph construction, respectively.
- We propose UniLR, a Unified Legal Retriever with two innovative components: KES introduces attention supervision to guide the retriever in focusing on key elements and GKA combines graph attention and convolution to aggregate meta information.
- Extensive experiments on multiple datasets of common legal tasks demonstrate that UniLR achieves the best retrieval performance and significantly enhances LLM capabilities. All data and code are publicly available [2].

## 2 Related Work

### 2.1 Legal Artificial Intelligence

In recent years, researchers have focused on using NLP technology to solve specific tasks in the legal field, such as charge prediction (Zhong et al., 2018; Yang et al., 2019; Xu et al., 2020; Yue et al., 2021a; Wu et al., 2022; Chalkidis et al., 2020), article recommendation (Chen et al., 2013; Raghav et al., 2016; Louis and Spanakis, 2022), case retrieval (Raghav et al., 2016; Shao et al., 2020; Li et al., 2023a), court view generation (Wu et al., 2020; Yue et al., 2021b; Li et al., 2024b; Liu et al., 2024), and legal question answering (Zhong et al., 2020b; Kien et al., 2020; Louis et al., 2024). Recently, with the development of LLMs, researchers have transformed various legal tasks into question-and-answer pairs to fine-tune the LLMs, hoping to build unified legal LLMs to solve problems (Cui et al., 2023a; Liu et al., 2023; Huang et al., 2023). However, due to the lack of domain knowledge, both universal LLMs and legal LLMs perform poorly on some legal tasks. Previous work enhances legal judgment prediction and court view generation through various forms of knowledge injection, such as charge definitions, document templates, and reasoning rules (Li et al., 2024a; Zhou et al., 2024a; Li et al., 2025). Because LLM can read text form knowledge directly, a promising approach is enhancing LLMs through retrieval. In the legal

[1]The detailed definitions and acquisition of "key elements" and "meta information" are provided in Sec. 3.2

[2]https://github.com/LIANG-star177/ULKR

domain, researchers have explored retrieval for legal cases and articles(Li et al., 2023a; Wang et al., 2019a), yet there lacks a unified retriever for legal knowledge.

### 2.2 LLMs and Retrieval

LLMs can learn the knowledge in the retrieved information, which has been validated on many LLMs, such as GPT-3 (Brown et al., 2020), GPT-Neo (Black et al., 2021), and LLaMA (Touvron et al., 2023). In traditional retrieval methods, researchers generally use the BM25 algorithm or dense retrievers based on pre-trained models to retrieve from the training set (Liu et al., 2022; Rubin et al., 2022; Izacard et al., 2022). Recently, researchers have studied the harmonious integration of large models and retrieval (Li et al., 2023b; Luo et al., 2023; Lv et al., 2025). Ge et al. (2023) retrieves knowledge from diverse sources to improve query representation. Ma et al. (2023) uses reinforcement learning to train a query optimizer for semantic alignment. Wang et al. (2023) incorporates contextual recall in pre-training to familiarize models with RAG patterns. However, compared to general retrieval, legal knowledge presents diverse formats and potential confusion, requiring domain-specific solutions. This paper proposes a unified legal knowledge retriever to enhance LLMs in legal tasks.

## 3 Methodology

In this section, we first formalize the RAG problem for utilizing LLMs in legal tasks and then introduce our UniLR by detailing two parts: Knowledge Mining and Model Architecture. The overall approach of UniLR is illustrated in Fig. 2.

### 3.1 Problem Formulation

In this paper, we aim to develop a unified retriever that can perform different legal retrieval tasks. We first formulate the problem as follows:

Given a test example $(q, y)$ in task $t$, where $q$ is a legal query, $y$ is the true label, and the specific knowledge corpus $\mathcal{C}_t = \{c_1, ..., c_{n_c}\}$ consisting of $n_c$ knowledge entries, the probability of the LLM generating the target $y$ based on the query $q$ is defined as follows:

$$p(y|q) = LLM(y|T(c_1, ..., c_k; q)) \quad (1)$$

Here, $k$ is the number of retrieved knowledge entries, $T$ is the template for packaging the retrieved knowledge and the query. We expect the retriever to seamlessly adapt to task transitions.

### 3.2 Knowledge Mining

#### 3.2.1 Key elements extraction

According to previous work, legal knowledge often describes one or more events (Shen et al., 2020; Li et al., 2020). This paper's key elements correspond to the subjects, objects, and triggers that constitute a legal event. LEVEN (Yao et al., 2022) is a specialized legal event dataset, which groups legal facts into 108 event types, annotating them based on subjects, objects, and trigger words. We observe that while the event prediction model trained on LEVEN places high attention on key elements, it is not well-suited for retrieval tasks. Therefore, we distill the expertise of DMBERT (Wang et al., 2019b), which has the best event prediction performance in the LEVEN.

Specially, given a piece of legal knowledge entry $c$, we input it into the DMBERT to obtain the attention matrix $A$ of the final layer. Then the attention distribution for the $i$-th token $\hat{p}_i$ is calculated:

$$\hat{p}_i = \text{softmax}(A_i) = \frac{\sum_{j=1}^{l_c} \exp(A_{ij})}{\sum_{i=1}^{l_c}\sum_{j=1}^{l_c} \exp(A_{ij})} \quad (2)$$

Here, $l_c$ is the knowledge length. Finally, we obtain the attention distribution $\hat{P} = \{\hat{p}_1, ..., \hat{p}_{l_c}\}$ and construct a key element dictionary $\mathcal{D} = \{(c_1 : \hat{P}_1), ..., (c_{n_c} : \hat{P}_{n_c})\}$, where knowledge entries serve as keys and attention distributions as values. In this way, key elements from all corpus are extracted in the form of attention.

#### 3.2.2 Knowledge graph construction

A piece of legal knowledge is intricately linked to additional information, such as other knowledge, hierarchical structures, and legal events, which we refer to as its "meta information". To represent these associations comprehensively, we construct a heterogeneous graph as illustrated in Fig. 3.

Specifically, the heterogeneous graph $\mathcal{G} = (N, E)$ contains different types of nodes and edges. Firstly, we categorize the nodes into three types: (i) Knowledge nodes, $N_c = \{n_i^c\}$ represent individual pieces of legal knowledge. (ii) Hierarchy nodes, $N_h = \{n_i^h\}$ represent the hierarchical structure of knowledge within the tree-like Chinese legal system (Qin et al., 2024) (e.g., "Article 273" belongs to "criminal law/property infringement/theft"). (iii) Legal event nodes, $N_e = \{n_i^e\}$, represent the legal

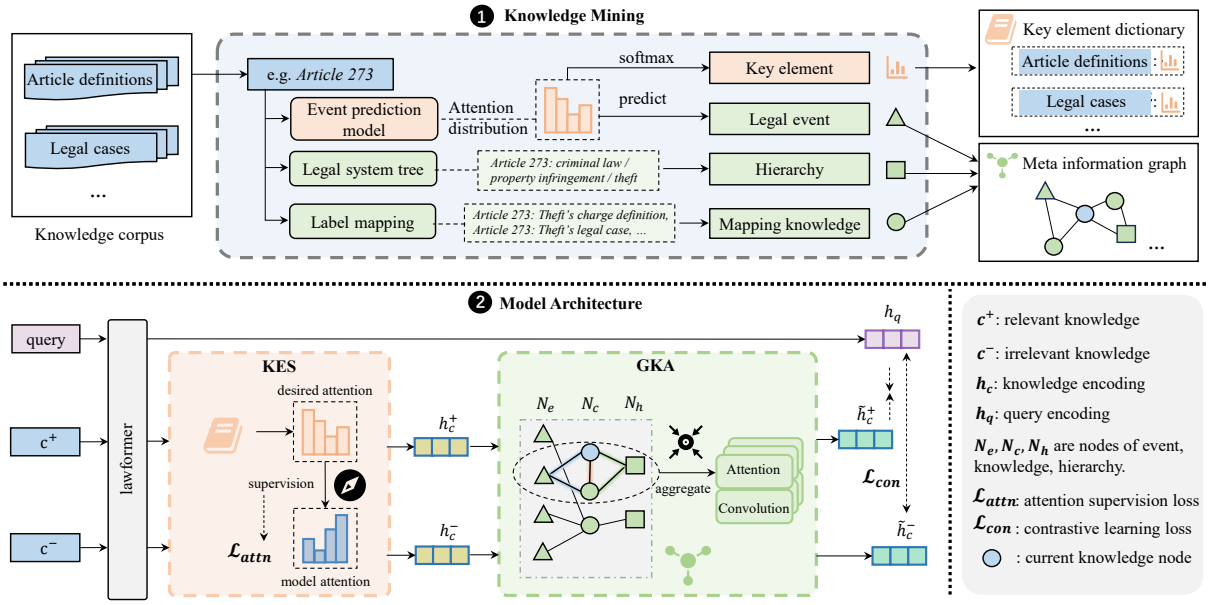



Figure 2: The overview of our UniLR. (1) is the knowledge mining process. We leverage the expertise of a legal event prediction model (Yao et al., 2022) to extract key elements into a dictionary and explore meta information to construct a heterogeneous graph. (2) is the model architecture. When encoding knowledge, KES guides the model to focus on key elements through attention supervision. GKA aggregates meta information to further enrich knowledge representation.

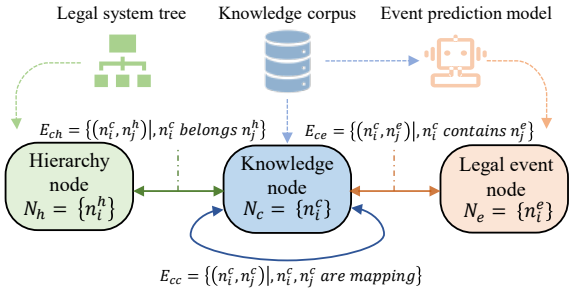



Figure 3: The definition of nodes and edges.

event labels predicted by DMBERT. We use the event label definitions from LEVEN. For node feature initialization, we use Lawformer (Xiao et al., 2021), a BERT variant pre-trained on extensive legal texts to obtain embedding.

The edges in the graph are defined based on specific connection rules: (i) If a knowledge node $n_i^c$ belongs to a hierarchy, whose node is $n_j^h$, an edge is created and belongs to $E_{ch}$. (ii) If a knowledge node $n_i^c$ contains the legal event, whose node is $n_j^e$, an edge is created and belongs to $E_{ce}$. (iii) If two knowledge nodes $n_i^c$ and $n_j^c$ have the same charge label, we think they are mapping, an edge is created and belongs to $E_{cc}$.

## 3.3 Model Architecture

### 3.3.1 Key Element Supervisor

To guide the retriever to focus on key elements when encoding knowledge, we introduce an attention supervision method. For the knowledge $c$, we input it into Lawformer. According to Eq. 2, we extract the model's attention distribution of each token and obtain $P = \{p_1, ..., p_{l_c}\}$. Next, we refer to the key element dictionary $\mathcal{D}$ to obtain the desired attention distribution $\hat{P}$ corresponding to knowledge $c$. To align the model's attention with the desired distribution, we introduce a loss function $\mathcal{L}_{attn}$, which minimizes the Kullback-Leibler (KL) divergence between the two distributions:

$$\mathcal{L}_{attn} = \sum_{i=1}^{l_c} KL(\hat{P}_i \parallel P_i) \quad (3)$$

Here, $l_c$ denotes the number of tokens in the knowledge. Then we obtain the hidden state of the final layer by applying the aligned attention distribution, denoted as $h^c = [h_1; ...; h_{l_c}] \in \mathbb{R}^{l_c \times d}$, where $d$ is the hidden dimension. By employing this approach, we effectively distill the expertise of DMBERT into the retriever, guiding the model to focus on key legal elements rather than the format.

### 3.3.2 Graph-based Knowledge Augmenter

To further enrich knowledge representation, we design a graph-based approach to aggregate meta

information. First, we extract the subgraph relevant to the knowledge from the entire graph $\mathcal{G}$. Specifically, for a piece of knowledge $c$, we identify the neighboring nodes to form a node set $N'$, which consists of $n'$ nodes. The edges connecting these nodes form an edge set $E'$. From this, we obtain a subgraph $\mathcal{G}' = \{N', E'\}$ relevant to the current knowledge.

Inspired by Guo et al. (2019), we employ graph attention to facilitate node interactions. Given GKA comprising $L$ layers, the node representation at the $l$-th layer is denoted as $H^{(l)} = [h_1^{(l)}; ...; h_{n'}^{(l)}]$. To ensure that a node is not influenced by nodes that are not directly connected, we prepare a mask for each node, which defines the index set $I_i$ of connected nodes for the $i$-th node. We then apply the Multi-Head Attention mechanism to learn multiple sets of attention weights, with the attention matrix for the $m$-th head given as follows:

$$I_i = \{j \mid i, j \in N', (n_i \leftrightarrow n_j)\} \tag{4}$$

$$\tilde{A}^{(m)} = softmax(\frac{(h_i^{(l)} W_i^Q) \cdot (h_i^{(l)} W_i^K)^T}{\sqrt{d}}), i \in I_i \tag{5}$$

Here, $W_i^Q \in \mathbb{R}^{d \times d}$ and $W_i^K \in \mathbb{R}^{d \times d}$ are learnable weight matrices used for linear transformations, $\leftrightarrow$ denotes existing edge between nodes.

To capture the high-order interactions, we design the heterogeneous graph convolution that considers different edge types. For the $i$-th node, we concatenate its encoding with the output from previous layers to serve as the input for the next graph layer, $g_i^{(l)} = [h^c; h_i^{(1)}; ...; h_i^{(l)}]$. The convolutional operation is then performed as follows:

$$h_i^{(l+1)} = \|_{m=1}^{M} \sigma(\sum_{j=1}^{n'} \tilde{A}_{ij}^{(k)} \cdot W_m^{(l)} \cdot g_j^{(l)} \cdot t_{ij} + b_m^{(l)}) \tag{6}$$

Here, $M$ is the number of attention heads, $W_m^{(l)}$ is learnable weights for node features, $t_{ij}$ is a one-hot vector indicating the edge type between $n_i$ and $n_j$, and $\sigma$ is the activation function. After propagating through the graph network, we aggregate the node features in the final layer:

$$H^{(L)} = MaxPooling[h_1^{(L)}; ...; h_{n'}^{(L)}] \in \mathbb{R}^{T' \times d} \tag{7}$$

To ensure stable gradient propagation, we flatten the node feature to $H_{flat}^{(L)} \in \mathbb{R}^{T'd}$, then concatenate it with the initial feature $h^c$ and reduce the dimension via a projection layer:

$$\tilde{h}^c = RELU(W \cdot [H_{flat}^{(L)}; h^c] + b) \in \mathbb{R}^d \tag{8}$$

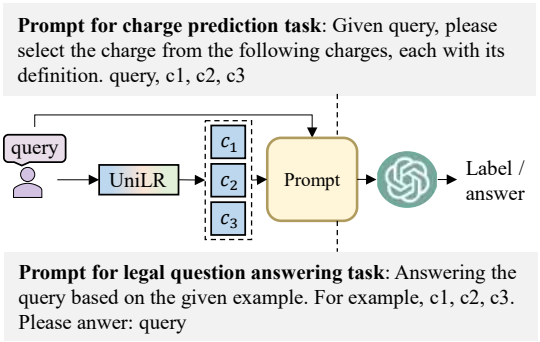



Figure 4: The process of performing legal tasks using UniLR and LLM in a RAG framework.

$W \in \mathbb{R}^{(T'd+d) \times d}$ is the weight matrix, and $b \in \mathbb{R}^{d \times 1}$ is the bias vector.

### 3.4 Contrastive Learning

Inspired by Gao et al. (2021), we use contrastive learning to concurrently train our model across multiple legal retrieval tasks to achieve uniformity.

For each query $q$, we obtain positive samples $Pos$ by selecting knowledge with the same charge label. For example, for a *Theft* case, we choose the charge definitions of *Theft*, *Article 273*, and other cases involving *Theft*. Negative samples $Neg$ are obtained through random sampling and hard negative sampling. When random sampling, we randomly select knowledge with different charge labels. When hard negative sampling, we select knowledge closely related within the same legal hierarchy, providing more challenging contrasts. Then, we permute positive and negative knowledge pairs and train our model by the following loss:

$$\mathcal{L}_{con} = -\sum_{\tilde{h}_c^+ \in Pos} log \frac{exp(sim(h_q, \tilde{h}_c^+)/\tau)}{\sum_{\tilde{h}_c^- \in Neg} exp(sim(h_q, \tilde{h}_c^-)/\tau)} \tag{9}$$

$h_q, \tilde{h}_c^+, \tilde{h}_c^-$ are representations of query, positive samples, and negative samples from UniLR, $sim$ calculates the cosine similarity, and $\tau$ is the temperature that adjusts the contrastive strength.

### 3.5 Training and Inference

In the training process, we combine data from multiple tasks for joint training. We employ contrastive learning loss to bring the relevant knowledge representation closer to the query, and attention distribution loss to emphasize the key elements. Therefore, the overall training objective of UniLR is to minimize the following combined loss:

$$\mathcal{L}_{total} = \mathcal{L}_{con} + \lambda \mathcal{L}_{attn} \tag{10}$$

| Types | Article | Charge | CVG | QA |
|---|---|---|---|---|
| **# Train** | 70973 | 70973 | 97863 | 8451 |
| **# Test** | 1683 | 1683 | 2039 | 387 |
| **# Knowledge corpus** | 483 | 452 | 6124 | 1248 |
| **Avg. # query length** | 440 | 440 | 558 | 45 |
| **Avg. # knowledge length** | 113 | 160 | 571 | 401 |

Table 1: Dataset details and knowledge base.

In the inference process, we employ UniLR to encode query and knowledge. Then, the cosine similarity is used to compute their relevance scores. The knowledge with the highest relevance scores is selected to assist the LLM in solving legal tasks. As shown in Fig. 4, for the charge prediction task, UniLR retrieves several label definitions and LLM selects the final label from them. A similar process is used for the law article prediction task. For the legal question answering task, we retrieve similar QA pairs and employ few-shot learning to guide LLM in producing the final answer. A similar process is used for the court view generation task.

# 4 Experiments

## 4.1 Dataset details and knowledge base

We conduct experiments on multiple common legal tasks: law article prediction (Article), charge prediction (Charge), court view generation (CVG), and legal question answering (QA). The dataset details and corresponding knowledge base are shown in Tab.1. **CAIL2018-Article** and **CAIL2018-Charge** are from the Chinese AI and Law challenge legal judgment prediction dataset [3] (Xiao et al., 2018). To align with prior works (Xu et al., 2020; Yue et al., 2021a), we filter out data with multiple articles and charges. In these datasets, the query is a fact description, and the task is to predict article and charge labels, respectively. **LAIC2021-CVG** is the court view generation dataset from Legal AI challenge [4], where the query is a fact description and the task is to generate the court view. **Lawzhidao-QA** is the criminal question-answering dataset selected from Baidu Legal Question Answering competition[5], where the query is a legal question and the task is to generate the answer. The details of the knowledge source can be found in Appendix A.

[3]http://cail.cipsc.org.cn/index.html
[4]https://data.court.gov.cn/pages/laic2021.html
[5]https://aistudio.baidu.com/datasetdetail/95693

## 4.2 Baselines

We implement three parts of baselines for comprehensive comparison and set retrieved knowledge number $k = 3$, with experimental settings detailed in Appendix B. We also conduct further experiments in Appendix C, including performance variation with different $k$ values and pre-trained models and runtime analysis.

**Traditional task-specific methods**. For prediction tasks, **TopJudge** (Zhong et al., 2018) captures topological dependencies among the subtasks in legal judgment prediction. **LADAN** (Xu et al., 2020) uses graph distillation for distinguishing charges and law articles. **NeurJudge** (Yue et al., 2021a) splits fact descriptions using intermediate subtask results for prediction. For generation tasks, **BART** (Lewis et al., 2019) is a widely used bidirectional autoregressive Transformer model. **T5** (Raffel et al., 2020) is a transformer architecture model that follows a text-to-text transfer learning paradigm. **C3VG** (Yue et al., 2021b) follows a two-stage architecture which is extraction-generation.

**LLM methods**. For LLMs, we use **GPT4** (Achiam et al., 2023), **LLaMA-3** (Cui et al., 2023b), **GLM4** (Zeng et al., 2024). For legal LLMs, we use **LexiLaw** [6], **LaWGPT** (Zhou et al., 2024b).

**LLM with retriever methods**. To evaluate retrieval enhancements, we implement the following retrieval methods with top-performing LLMs: **BM25** (Robertson and Zaragoza, 2009) is a retrieval model based on term frequency and document length. **Contriever** (Izacard et al., 2022) is a dense embedding model trained via contrastive learning. **LED** (Zhang et al., 2023) enhances dense retrieval by aligning embeddings with lexicon-aware representations through weakened knowledge distillation. **SAILER** (Li et al., 2023a) is a legal case retriever that incorporates structural information and legal rules.

## 4.3 Evaluation Metrics

### 4.3.1 Retrieval evaluation

We use Hit@k (Norouzi et al., 2014) as the evaluation metric. if there is relevant knowledge among the $k$ knowledge retrieved, it is considered successful retrieval. Considering that the relevant knowledge is not unique when retrieving precedents, we

[6]https://github.com/CSHaitao/LexiLaw

| Methods | CAIL2018-Article | | | | | CAIL2018-Charge | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Ma-P | Ma-R | Ma-F | Acc | Hit@k | Ma-P | Ma-R | Ma-F | Acc | Hit@k |
| *Traditional task-specific methods* | | | | | | | | | | |
| **TopJudge** | 74.49 | 66.26 | 68.68 | 80.85 | - | 74.43 | 68.37 | 70.41 | 78.42 | - |
| **LADAN** | 75.61 | 70.29 | 70.46 | 80.92 | - | 75.36 | 70.02 | 71.19 | 79.45 | - |
| **NeurJudge** | 75.16 | 72.01 | 72.26 | **81.94** | - | 75.76 | 71.24 | 71.59 | 80.31 | - |
| *LLM methods* | | | | | | | | | | |
| **GPT4** | 14.76 | 13.46 | 13.47 | 14.68 | - | 45.97 | 35.61 | 34.38 | 46.94 | - |
| **LLaMA-3** | 15.18 | 12.43 | 13.19 | 14.00 | - | 40.51 | 31.88 | 33.09 | 42.00 | - |
| **GLM4** | 9.42 | 7.47 | 6.57 | 14.00 | - | 47.17 | 35.27 | 37.68 | 41.33 | - |
| **LexiLaw** | 12.27 | 6.54 | 7.41 | 9.33 | - | 36.59 | 32.81 | 37.73 | 42.33 | - |
| **LaWGPT** | 14.49 | 12.17 | 12.64 | 16.35 | - | 32.21 | 28.81 | 34.11 | 43.37 | - |
| *Best LLM (GPT4) with retriever methods* | | | | | | | | | | |
| **+ BM25** | 31.90 | 25.58 | 26.70 | 36.82 | 71.13 | 52.63 | 42.87 | 46.17 | 50.75 | 73.28 |
| **+ Contriever** | 72.35 | 63.41 | 67.83 | 73.76 | 75.06 | 73.10 | 65.90 | 66.09 | 71.37 | 81.58 |
| **+ LED** | 70.73 | 63.24 | 67.97 | 74.54 | 78.31 | 74.41 | 66.79 | 68.42 | 69.62 | 82.25 |
| **+ SAILER** | 73.75 | 65.85 | 69.76 | 77.91 | 80.70 | 74.76 | 68.95 | 69.70 | 76.75 | 83.90 |
| **+ UniLR** | **77.24** | **72.73** | **72.32** | 80.98 | **84.88** | **76.98** | **73.11** | **72.34** | **80.36** | **85.27** |

Table 2: The performance of article prediction task and charge prediction task. The best is **bolded**, the second best is underlined, and the gray-shaded metric only measures retriever performance when $k = 3$.

| Methods | LAIC2021-CVG | | | | | | Lawzhidao-QA | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | B-1 | B-2 | B-N | R-L | Hit@k | R-p@k | B-1 | B-2 | B-N | R-L | Hit@k | R-p@k |
| *Traditional task-specific methods* | | | | | | | | | | | | |
| **Bart** | 61.01 | 52.01 | 48.97 | 56.95 | - | - | 37.58 | 23.71 | 20.74 | 21.93 | - | - |
| **T5** | 61.24 | 51.24 | 47.68 | 58.09 | - | - | 36.32 | 21.45 | 20.24 | 18.31 | - | - |
| **C3VG** | 63.35 | 52.70 | 49.30 | 60.71 | - | - | - | - | - | - | - | - |
| *LLM methods* | | | | | | | | | | | | |
| **GPT4** | 41.23 | 26.90 | 23.06 | 27.46 | - | - | 11.98 | 7.63 | 5.73 | 14.38 | - | - |
| **LLaMA-3** | 35.98 | 19.74 | 18.01 | 23.17 | - | - | 20.70 | 15.27 | 14.10 | 18.39 | - | - |
| **GLM4** | 42.15 | 28.01 | 22.32 | 32.31 | - | - | 33.02 | 20.10 | 15.43 | 20.73 | - | - |
| **LexiLaw** | 39.30 | 23.26 | 20.59 | 23.91 | - | - | 28.99 | 16.56 | 14.42 | 19.23 | - | - |
| **LaWGPT** | 15.17 | 10.03 | 8.59 | 12.85 | - | - | 24.32 | 16.10 | 14.22 | 17.78 | - | - |
| *Best LLM (GLM4) with retriever methods* | | | | | | | | | | | | |
| **+ BM25** | 53.47 | 46.53 | 44.46 | 58.30 | 83.20 | 65.43 | 39.04 | 28.24 | 24.49 | 30.72 | 59.63 | 37.47 |
| **+ Contriever** | 57.35 | 51.07 | 48.98 | 63.51 | 87.50 | 77.32 | 42.26 | 30.38 | 26.42 | 32.95 | 65.22 | 55.10 |
| **+ LED** | 57.49 | 51.78 | 48.84 | 63.52 | 87.88 | 79.46 | 43.80 | 32.45 | 27.54 | 32.87 | 66.78 | 55.73 |
| **+ SAILER** | 59.06 | 53.02 | 50.98 | 65.05 | 90.97 | 87.86 | 45.00 | 34.93 | 30.82 | 36.26 | 70.43 | 60.73 |
| **+ UniLR** | **63.94** | **58.03** | **55.81** | **72.68** | **91.41** | **90.23** | **48.38** | **37.13** | **33.27** | **40.87** | **75.90** | **64.10** |

Table 3: The performance of court view generation task and legal question answering task. The best is **bolded**, the second best is underlined, and the gray-shaded metric measures retriever performance when $k = 3$.

select R-p@k (Chen et al., 2023) to do further evaluation. It is calculated as $r/R$, where $R$ is the number of retrieved knowledge and $r$ is the number of relevant knowledge.

#### 4.3.2 Task performance evaluation

For the prediction tasks, we employ macro precision (Ma-P), macro recall (Ma-R), macro f1 score (Ma-F), and accuracy (Acc). For the generation task, we use BLEU-1 (B-1), BLEU-2 (B-2), BLEU-N (B-N, the average value of BLEU-1 $\sim$4), and ROUGE-L (R-L) (Lin, 2004). To comprehensively test the generation results, we conduct the human evaluation, with details provided in Appendix D.

### 4.4 Experimental Results

#### 4.4.1 The performance of prediction tasks

From Tab.2, we can conclude that: (1) Traditional task-specific methods excel over LLMs. This is because LLMs are generative, selecting a token from a vast vocabulary in decoding. (2) Legal LLMs underperform universal LLMs like GPT4, possibly due to a decrease in in-context learning ability during fine-tuning. (3) In retrieval evaluation, our UniLR outperforms the best baseline SAILER by 5.18% in Hit@k for article prediction and 1.63% for charge prediction. This indicates the effectiveness of introducing key elements and meta information. (4)In prediction evaluation, UniLR significantly improves GPT4's perfor-

| Methods | CAIL2018-Article | | | | | CAIL2018-Charge | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Ma-P | Ma-R | Ma-F | Acc | Hit@k | Ma-P | Ma-R | Ma-F | Acc | Hit@k |
| **UniLR** | **77.24** | **72.73** | **72.32** | **80.98** | **84.88** | **76.98** | **73.11** | **72.34** | **80.36** | **85.27** |
| **w/o KES** | 74.15 | 71.31 | 70.76 | 78.87 | 81.33 | 75.25 | 69.02 | 69.83 | 77.89 | 81.68 |
| **w/o GKA** | 75.70 | 71.80 | 71.04 | 79.57 | 81.96 | 74.18 | 71.07 | 68.59 | 78.69 | 82.37 |
| *Ablation of meta information in GKA* | | | | | | | | | | |
| **w/o** $E_{cc}$ | 75.41 | 71.28 | 70.91 | 79.09 | 82.94 | 75.97 | 70.24 | 70.07 | 78.53 | 83.01 |
| **w/o** $E_{ce}$ | 76.45 | 72.64 | 72.07 | 80.57 | 84.14 | 76.42 | 72.26 | 71.83 | 79.59 | 84.58 |
| **w/o** $E_{ch}$ | 75.10 | 71.65 | 71.06 | 80.12 | 84.05 | 76.15 | 69.48 | 70.30 | 77.39 | 82.94 |

Table 4: Ablation experiment of the best performing LLM with UniLR in prediction tasks.

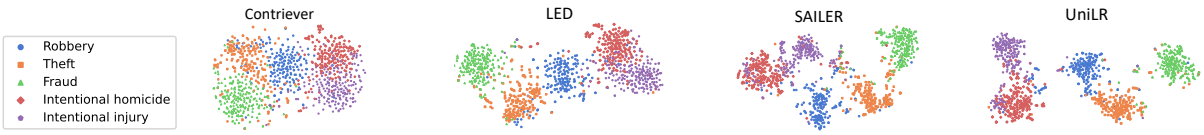



Figure 5: The t-SNE plots of legal cases.

| Methods | CVG | | QA | |
|---|---|---|---|---|
| | Flu. | Rat. | Flu. | Rat. |
| **BART** | 4.44 | 3.64 | 2.18 | 2.52 |
| **GLM4** | 3.37 | 2.71 | 3.09 | 2.91 |
| **GLM4+SAILER** | 4.46 | 3.93 | 3.95 | 3.47 |
| **GLM4+UniLR** | **4.62** | **4.04** | **4.28** | **3.99** |

Table 5: Human evaluation on legal generation tasks.

mance, surpassing all baselines, including the traditional SOTA method. This suggests that UniLR effectively bridges the gap between LLMs and legal tasks.

#### 4.4.2 The performance of generation tasks

From Tab.3 and Tab.5, we have the following observations: (1) LLM methods show poor performance, indicating their tendency to generate imaginative outputs are unsuitable for the precision required in the legal domain. (2) Many LLM with retriever methods surpass traditional task-specific methods emphasizing the importance of knowledge. (3) GLM4 surpasses GPT4, suggesting that its training involved more Chinese legal documents. (4) UniLR achieves top retrieval performance and significantly boosts GLM4's downstream task performance. For example, in legal QA, it outperforms the second-best baseline by 7.77% in Hit@k and 12.71% in R-L. (5) In human evaluation, UniLR excels in Fluency and Rationality, aligning with automatic evaluations. Additionally, BART performs well in CVG but lags in QA, while GLM4 shows the opposite trend due to the structured nature of court views versus the flexibility of QA tasks.

#### 4.4.3 Ablation study

We conduct ablation experiments on the GPT4+UniLR method in two prediction tasks, as shown in Tab. 4. (1) **w/o KES** eliminates the key elements supervision, relying solely on Lawformer encoding. We find a significant performance degradation, demonstrating the importance of focusing on key elements. (2) **w/o GKA** removes the graph-based knowledge augmenter. Noticeable performance degradation indicates that aggregating meta information can effectively capture a comprehensive knowledge representation. (3) We conduct ablation experiments on GKA's associations, including mapping knowledge, legal events, and hierarchy. Removing associations between knowledge results in the most significant performance decline.

### 4.5 Visual Analysis

Following Li et al. (2023a), to explore UniLR's distinguish ability for legal knowledge representation, we select 5000 illegal facts from CAIL2018 that involve confusing charges and visualize their encoding by using different retriever methods. Specifically, the charges included *Robbery, Theft, Snatch, Intentional injury, Intentional homicide*, each with 1000 cases. As observed from Fig. 5, UniLR significantly increases the distance between encoded illegal facts related to different charges, demonstrating its strongest ability to distinguish between confusing knowledge. Case Study in Appendix E also confirms the effectiveness of UniLR.

## 5 Conclusion and Future Work

In conclusion, we address the challenges in enhancing LLMs in multiple legal tasks by introducing a Unified Legal Knowledge Retriever (UniLR). By incorporating key elements and meta information, UniLR significantly alleviates disparities in multiple retrieval tasks and reduces knowledge confusion. Extensive experiments on multiple common legal tasks demonstrate that UniLR outperforms state-of-the-art retrieval methods and significantly enhances the performance of LLMs in legal applications. In the future, to better explore Retrieval-Augmented Generation (RAG) in legal tasks, we aim to make efforts in two directions: (1) Training an LLM capable of flexibly utilizing knowledge obtained from retrievers. (2) Leveraging feedback from the LLM to train a retriever that is better suited for LLM.

## 6 Ethical Issue Discussion

Legal AI has benefited from the emergence of LLMs, but it's a sensitive technology that demands ethical considerations. Our UniLR is designed to enhance LLMs in legal tasks, mitigating the risk of factual errors to some extent. However, even minor inaccuracies could have significant consequences. Our goal is to provide suggestions to judges rather than replace them. In practice, human judges should be the final safeguard to protect justice and fairness. Although our method demonstrates promising results on legal task datasets, it does not imply that it can endow LLMs with human-like empathy, experience, and intuition. It is necessary to prevent misuse. Additionally, given that the model retrieves knowledge from the external corpus, ensuring the quality and fairness of the corpus is crucial.

## 7 Limitations

In this section, we discuss the limitations of our works as follows:

- We do not pretrain or fine-tune LLMs for the RAG process. Training an LLM capable of flexibly utilizing knowledge obtained from retrievers, may produce better output.
- We validate the effectiveness of designing a legal retriever to meet LLM needs. Exploring the application of such retrievers in other fields like medicine and education is worth considering.
- Our research is conducted on the Chinese legal system. We are also very interested in exploring the generalization of our methods to other languages.

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## A Knowledge Source.

We collect multiple legal knowledge sources to construct the knowledge base. In our UniLR, they are further processed together into a legal essence dictionary and a heterogeneous knowledge metagraph.

(1) **Article definitions**: are detailed content about all criminal law articles. It assists in the article prediction task. (2) **Charge definitions**: are specific definitions for all criminal charges from the Criminal Law, formatted similarly to Xu et al. (2020). It assists in the charge prediction task. (3) **CVG cases**: are selected from the training set of LAIC2021 in a balanced way. It contains fact descriptions and corresponding court views. It assists in the court view generation task. (4) **QA cases**: are similarly selected from the training set of Lawzhidao-QA dataset. It assists in the legal QA task. (5) **Legal system tree**: is extracted from the Chinese Criminal Law, which has a tree-like structure comprising chapters and specific offenses. Each knowledge can be brought into a leaf node in the hierarchical structure, utilizing paths from the root to the leaf. (6) **Legal events**: are from the legal event detection dataset LEVEN (Yao et al., 2022). We focus on charge-oriented events and leverage their definitions.

# B Experiment Settings

All training and inference were conducted on 2 NVIDIA Tesla A100 GPUs. We rerun the experiments five times with different random seeds and report the average. We also use the Fisher randomization test to ensure the significance of the results.

## B.1 Retrieval setting

We set the maximum length for input query and each knowledge to 512. For the BM25 algorithm, we set $k1$ to 1.5 and $b$ to 0.75. Other retrieval methods utilizing pre-trained models (Contriever, LED, SAILER, UniLR) leverage the Lawformer (Xiao et al., 2021) as the pre-trained embedding model, followed by fine-tuning. To ensure fairness, the dense retrieval baselines are jointly fine-tuned on multiple legal retrieval tasks, similar to UniLR. In the fine-tuning, all retrievers are trained using the Adam optimizer (Kingma and Ba, 2017) with a learning rate of 1e-6 for 50 epochs. In our UniLR, we set the GKA layers to 2, and the hyperparameter controlling loss $\lambda$ to the best-performing value of 0.2. In the reference of retrieval, we use top-k selection to select the final knowledge and set k=3. Referring to Langhain Chatchat [7], we use FAISS to pre-vectorize the knowledge base, allowing for quick computation of relevance scores with the query during inference.

[7]https://github.com/chatchat-space/Langchain-Chatchat

## B.2 Legal tasks setting

For traditional methods, they are trained and inferred according to their original papers. For all LLMs, we set the inference temperature to 0.8 and the maximum output length to 512. Regarding LLM with retrieval methods, since they require concatenating the query and knowledge, we set the maximum input length to 2048.

# C Further Experiments

## C.1 The performance variation with different k values

In this paper, we set the number of retrieved knowledge number $k = 3$ for each query. We explore the impact of different $k$ values on the performance of multiple legal retrieval tasks, testing experimental results for $k$ ranging from 1 to 5. As shown Tab. 6, when $k \leq 3$, the performance gain from adding retrieved knowledge number is relatively significant. However, further increasing $k$ brings minimal improvement. Moreover, inference with the LLM consumes substantial computational resources. Considering the trade-off between performance gain and resource consumption, we chose $k = 3$ for the main experiment.

## C.2 Sensitivity analysis of the combined loss hyperparameter

To evaluate the effect of the hyperparameter $\lambda$ in the combined loss function, we conducted experiments on four datasets by varying $\lambda$ within the range [0.1,0.4]. The results are shown in Tab. 7. The results indicate that when $\lambda < 0.2$, the attention supervision signal is insufficient, resulting in lower performance. Conversely, when $\lambda > 0.2$, the attention supervision component dominates the optimization objective and suppresses the primary task loss, leading to degraded performance. Optimal results are achieved when $\lambda = 0.2$, which provides a balanced trade-off between task-specific loss and attention supervision.

## C.3 The performance variation with different pre-trained model

To assess the robustness of our pipeline, we replaced Lawformer with BERT-base Chinese. We also set the retrieved knowledge number $k = 3$. The experimental results for law article prediction task and charge prediction task are shown in Tab. 8 and Tab. 9. The BERT pre-trained model was not specifically trained on legal texts, yet it did

| k | CAIL2018-Article | CAIL2018-Charge | LAIC2021-CVG | Lawzhidao-QA |
|---|---|---|---|---|
| 1 | 75.72 | 76.37 | 90.63 | 65.96 |
| 2 | 80.73 | 81.91 | 91.21 | 72.17 |
| 3 | 84.88 | 85.27 | 91.41 | 75.90 |
| 4 | 86.05 | 86.25 | 91.60 | 77.76 |
| 5 | 86.84 | 87.22 | 91.99 | 78.39 |

Table 6: Performance of Different $k$ Values

| $\lambda$ | CAIL2018-Article | CAIL2018-Charge | LAIC2021-CVG | Lawzhidao-QA |
|---|---|---|---|---|
| 0.1 | 83.78 | 84.11 | 90.41 | 73.93 |
| 0.2 | **84.88** | **85.27** | **91.41** | **75.90** |
| 0.3 | 84.71 | 83.59 | 91.54 | 74.11 |
| 0.4 | 84.40 | 82.49 | 90.78 | 73.20 |

Table 7: Performance with different values of $\lambda$ in the combined loss function.

| Methods | Ma-F | Acc | Hit@k |
|---|---|---|---|
| UniLR (BERT) | 71.84 | 80.73 | 82.84 |
| UniLR (Lawformer) | 72.32 | 80.98 | 84.88 |

Table 8: The performance of law article prediction task.

| Methods | Ma-F | Acc | Hit@k |
|---|---|---|---|
| UniLR (BERT) | 72.22 | 79.89 | 84.96 |
| UniLR (Lawformer) | 72.34 | 80.36 | 85.27 |

Table 9: The performance of charge prediction task.

not cause a significant performance drop in UniLR. These results demonstrate that UniLR maintains robust performance when replacing the pre-trained model.

### C.4 Runtime analysis

We further analyze the runtime as follows: During training, we used a distributed setup on a single machine with 2 A100 GPUs. Each GPU utilized 16,964MB of memory, with a batch size of 8. The model was trained for 50 epochs, completing in approximately 5.56 hours. For inference, the model uses FAISS to pre-store knowledge as a vector database, which is 44.4MB in size. During inference, the model occupies 4,356MB of GPU memory. Retrieving results from the model is quick, with the primary time consumption being LLMs generating answers based on the retrieved knowledge. On average, generating each response takes 1.53 seconds.

## D Human Evaluation Metric

We select baselines with good similarity performance to do human evaluation. We randomly select 200 samples and shuffle them to ensure fairness. We invite five annotators (10 Ph.D. students from the Law major) to evaluate every sample from two perspectives referencing true labels:

- **Fluency.** The annotators rate the fluency of generation texts on a scale of 1-5.
- **Rationality.** The annotator needs to score 1-5 on whether the answer to the question is reasonable.

When scoring on a scale of 1-5, they are required to provide integer scores, i.e., selecting from the range [1, 2, 3, 4, 5].

## E Case Study

We conduct the case study to demonstrate the application details of UniLR in legal prediction tasks. In Fig. 6, based on the fact description, the defendant was discovered stealing property and then engaged in violent behavior (forcibly dragging the victim), which constituted a robbery. UniLR correctly identified this and accurately predicted *Robbery*, while NeurJudge predicted theft based on the act of *Theft*.

We also provided a case for legal QA in Fig. 7, we use gray highlights to represent confusing semantics. Red highlights represent key elements, while green highlights represent meta information. It is observed that SAILER, influenced by the format *in order to* and the word *detain*, retrieves a *Detention* case. In reality, due to the defendant's intention for valuables and the violent actions, the charge has transformed into *Kidnapping*. UniLR focuses on key elements such as valuables and violence in the knowledge, and further enriches the knowledge information by associating the legal

| | | | | |
|---|---|---|---|---|
| **Fact Description** | At around 2 a.m. on August 4, 2015, the defendant, W, drove a tricycle to a foot bath shop in Cixi City and stole a white Apple 4S phone worth RMB 750 while the victim, L, was asleep. As W was escaping, L woke up and grabbed the tricycle. Despite knowing this, W accelerated, dragging L for about 20 meters and causing L to fall and sustain minor injuries. The stolen phone was recovered and returned to L. | | | |
| **Judgment** | | Ground truth | NeurJudge ✗ | UniLR ✓ |
| | Law articles | Article 263 | **Article 263** (Anyone who steals a significant amount of public or private property, commits theft multiple times, breaks into a residence to steal, carries a weapon during theft, or engages in pickpocketing shall be sentenced to up to 3 years of imprisonment, detention, or control, and fined..) | **Article 263** (Anyone who robs public or private property by violence, threat, or other means shall be sentenced to 3 to 10 years of imprisonment and fined) |
| | charges | Robbery | **Theft** (Definition: Theft is the act of secretly taking a significant amount of public or private property, or repeatedly secretly taking public or private property, with the intent of illegal possession.) | **Robbery** (Definition: Robbery is the act of unlawfully taking public or private property by force, threat, or other means with the intent of illegal possession.) |

Figure 6: A case study of article and charge prediction tasks.

| | | |
|---|---|---|
| **Query** | In order to quickly obtain valuables, defendant A detained victim B for an extended period and violently demanded cash. Is the sentence in this case serious? | |
| **Ground Truth** | According to Article 239 of the Criminal Law, the defendant's behavior seriously violates the individual's right to freedom and personal safety. Defendant A kidnap the victim for the purpose of forcing him to provide property, and he shall be sentenced to fixed-term imprisonment of not less than ten years or life imprisonment. | |
| **Retrieved Knowledge** | SAILER ✗ | UniLR ✓ |
| | **Query:** In order to seek revenge, defendant A illegally detained and insulted Victim B. How will he be sentenced?<br>**Answer:** According to the law, illegally detaining others shall be sentenced to fixed-term imprisonment of not more than three years, detention, public surveillance, or deprivation of political rights. | **Query:** Defendant A kidnapped and extorted Victim B 5 million yuan. The stolen money has been spent and A can not repay civil compensation. What is the verdict?<br>**Answer:** According to the law, kidnapping for ransom of 5 million yuan is suspected of kidnapping and should be sentenced to more than ten years in prison or life imprisonment.<br>**Legal event Kidnapping:** Using violent means to take hostages in exchange for benefits, the target is a person.<br>**Article knowledge:** Article 239, Anyone who kidnaps others with the purpose of extorting money or property shall be sentenced to at least ten years… |

Figure 7: A case study of Legal QA task.

event and the article. Ultimately, UniLR successfully retrieves the appropriate QA case of *Kidnapping*.